

Linear Algebra - 24 DS

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March 11, 2026

1 Problem Sheet - Matrices

1. Check whether the following matrix is singular:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

2. Determine if the matrix is singular:

$$B = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 0 & 6 \\ -2 & 1 & -3 \end{pmatrix}$$

3. For what value of k is the matrix singular?

$$C = \begin{pmatrix} k & 1 & 2 \\ 3 & k & 4 \\ 5 & 6 & k \end{pmatrix}$$

4. Find the inverse of the matrix using the adjoint method and verify $AA^{-1} = I$:

(a)

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$$

(b)

$$A = \begin{pmatrix} 2 & 3 & 1 \\ -1 & 2 & 0 \\ 4 & -2 & 1 \end{pmatrix}$$

5. Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 2 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Verify that $(AB)^{-1} = B^{-1}A^{-1}$.

6. For

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Show that $(PQ)^{-1} = Q^{-1}P^{-1}$.

7. A 3×3 matrix M is used in computer graphics to represent rotation about the z-axis by an angle θ :

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Show that this matrix is orthogonal ($R^{-1} = R^T$) and find its determinant. What does the determinant signify in transformation?

8. In solving a system $A\mathbf{x} = \mathbf{b}$ for 3 variables in a simulation, the matrix

$$A = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix}$$

appears. Is the matrix invertible? If yes, find A^{-1} and explain why invertibility is important in this context.

9. A symmetric matrix in machine learning covariance calculation is given by

$$S = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

Check if S is invertible. If so, find S^{-1} and verify it is also symmetric.

Determinant Splitting Property: If a row (or column) of a matrix A has entries that can be written as the sum of two numbers, then

$$\det(A) = \det(B) + \det(C),$$

where B and C are matrices identical to A except that the row (or column) in question has the first and second numbers of the sum respectively.

10. Let

$$A = \begin{pmatrix} 2+3 & 1 & 4 \\ 1 & 0 & 2 \\ 5 & 2 & 1 \end{pmatrix}.$$

Use the determinant splitting property to write $\det(A)$ as $\det(B) + \det(C)$, where B and C are matrices formed by splitting the first row. Verify by computing $\det(A)$ directly and comparing with $\det(B) + \det(C)$.

11. Consider the matrix

$$A = \begin{pmatrix} 4 & 1 & 2 \\ 0 & 1+2 & 3 \\ 1 & 0 & 3 \end{pmatrix}.$$

- Apply the splitting property to the second row (where the second entry is $1 + 2$) to express $\det(A)$ as $\det(B) + \det(C)$.
- Now, instead, apply the property to the third column if its entries were written as sums (e.g., $2 = 1 + 1$, $3 = 1 + 2$, $3 = 0 + 3$), and verify that the sum of determinants remains the same.
- Compute all determinants explicitly to check the property.